

The cone, the cylinder, the prism and the pyramid

Four families of solid, told apart by their cross-sections — and one volume rule that covers two of them.

LESSON 176–178

3 × 40 MIN

MATEMATIKA 6, §34

S7 8.4 3-D SHAPES

LESSON CLOCK

30'

30'

30'

25'

5

Prisms and the cross-section	30 min
The cylinder	30 min
Pyramids and the cone	30 min
The volume of a prism	25 min
Homework	5 min

LEARNING OBJECTIVES

- Describe a prism by its cross-section and count its parts.
- Describe a cylinder, a cone and a pyramid, naming their surfaces.
- Find the volume of a prism as base area times height.
- Find the volume and the curved surface of a cylinder.

TEXTBOOK

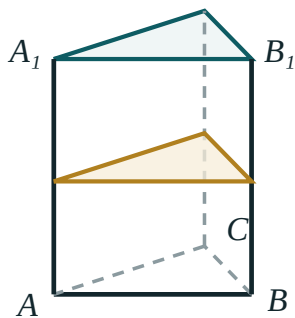
National	pp. 502–511
Cambridge	Stage 7, pp. 84–90
Workbook	Exercise 8.4

01

Explanation

Prisms and the cross-section

A prism has two identical ends joined by rectangles. Slice it anywhere across and the face you expose is always that same shape — its **cross-section**.



a section parallel to the base is congruent to it

every cross-section of a prism repeats the base

Cutting a prism across gives the same shape every time

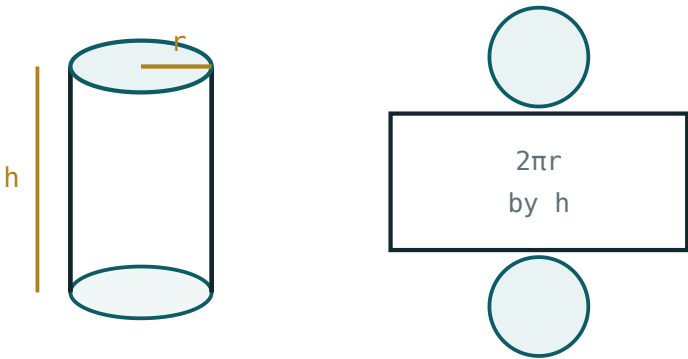
PRISM	CROSS-SECTION	FACES	EDGES	VERTICES
cuboid	rectangle	6	12	8
triangular prism	triangle	5	9	6
pentagonal prism	pentagon	7	15	10
hexagonal prism	hexagon	8	18	12

A CUBOID IS A PRISM, AND SO IS A CUBE

Nothing new has to be learnt for them: they are the case where the cross-section happens to be a rectangle or a square.

The cylinder

A cylinder is a prism whose cross-section is a circle — except that its “rectangles” have curled into a single curved surface.



$$S = 2\pi r^2 + 2\pi rh$$

— two circles and one rectangle

The net of a cylinder — two circles and one rectangle

PART	SHAPE	ITS MEASUREMENT
the two ends	circles of radius r	πr^2 each
the curved surface	a rectangle when unrolled	$2\pi r \times h$
the whole surface	both together	$2\pi r^2 + 2\pi rh$

The unrolled rectangle is as long as the circle it came from — that is why its width is the circumference $2\pi r$.

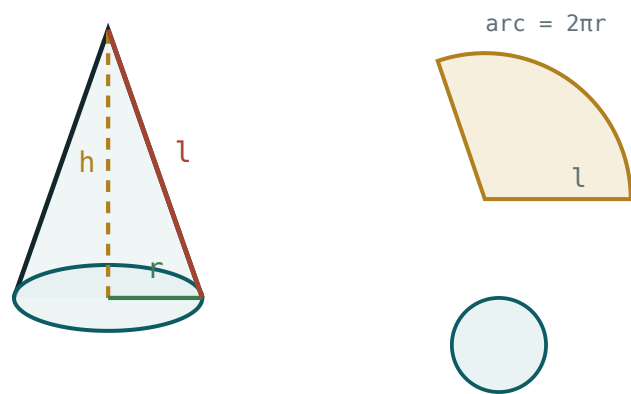
CYLINDER	END AREA	CURVED SURFACE	TOTAL SURFACE
$r = 5, h = 10$	78.5 cm^2	314 cm^2	471 cm^2
$r = 3, h = 8$	28.3 cm^2	150.7 cm^2	207.2 cm^2

TAKE $\pi \approx 3.14$ AND SAY SO

Every answer here is rounded. Writing “taking $\pi = 3.14$ ” beside the working is what makes 471 a correct answer rather than an approximate one.

Pyramids and the cone

A pyramid has one base and triangles rising to a single point, the **apex**. A cone is the same idea with a circular base and one curved surface.



$S = \pi r^2 + \pi rl$ – a circle and a sector
The net of a cone — a circle and a sector

SOLID	BASE	SIDES	FACES	EDGES	VERTICES
triangular pyramid	triangle	3 triangles	4	6	4
square pyramid	square	4 triangles	5	8	5
hexagonal pyramid	hexagon	6 triangles	7	12	7
cone	circle	one curved surface	1 flat	one curved edge	1 apex

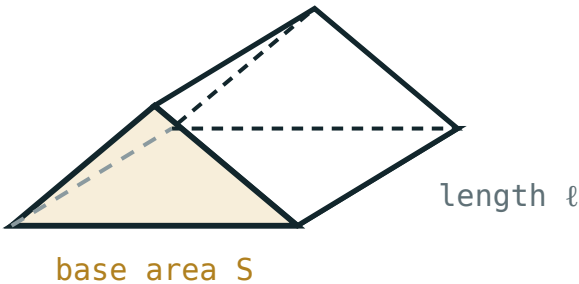
A PYRAMID HOLDS A THIRD OF THE PRISM ON THE SAME BASE

Pour sand from a pyramid into a prism of the same base and height and it fills exactly one third of it — and a cone fills a third of its cylinder. Grade 11 proves it; a jug of sand shows it today.

The volume of a prism

The cuboid rule $V = abc$ was really $(a \cdot b)$ — a base area — multiplied by the height. That is true of every prism, whatever its cross-section.

$V = \text{base area} \times \text{height}$



$V = S \cdot \ell$

Base area times height — the rule for every prism

PRISM	CROSS-SECTION AREA	LENGTH	VOLUME
triangular, base 6 by height 4	12 cm ²	10 cm	120 cm ³
trapezium ended	20 cm ²	8 cm	160 cm ³
cylinder, $r = 5$	78.5 cm ²	10 cm	785 cm ³
cylinder, $r = 3$	28.3 cm ²	8 cm	226 cm ³

ONE RULE, FOUR SOLIDS

Find the area of the end, multiply by the length. The only work that changes from one prism to the next is the area of the cross-section — a triangle, a trapezium, or a circle.

02 Worked examples

EXAMPLE 1 A triangular prism has a cross-section with base 6 cm and height 4 cm, and is 10 cm long. Find its volume.

① Cross-section $\frac{1}{2} \cdot 6 \cdot 4 = 12 \text{ cm}^2$.

② $12 \cdot 10$.

③ $= 120 \text{ cm}^3$.

Area in cm^2 times a length ✓

ANSWER

120 cm^3

EXAMPLE 2 A cylinder has radius 5 cm and height 10 cm. Find its volume, taking $\pi = 3.14$.

① End area $3.14 \cdot 25 = 78.5 \text{ cm}^2$.

② $78.5 \cdot 10$.

③ $= 785 \text{ cm}^3$.

Taking $\pi = 3.14$ ✓

ANSWER

785 cm^3

EXAMPLE 3 A prism has volume 240 cm^3 and length 8 cm . Find the area of its cross-section.

① $V = \text{area} \times 8.$

② $240 \div 8.$

③ $= 30 \text{ cm}^2.$
 $\text{cm}^3 \div \text{cm} = \text{cm}^2 \checkmark$

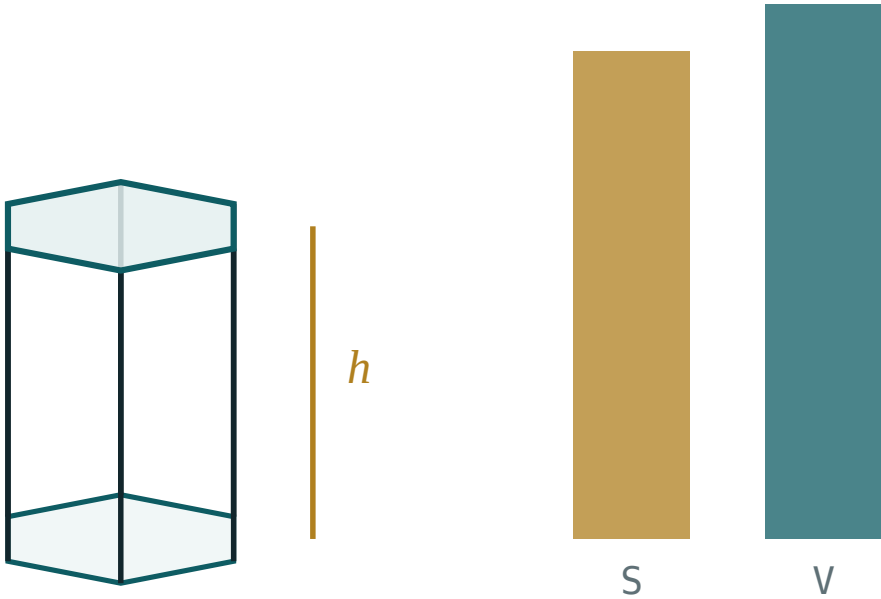
ANSWER

30 cm^2

03 Interactive model

Set the base of the model to four sides and then to many; as the polygon rounds off into a circle the prism becomes a cylinder in front of the class.

Base area times height



base area B 41.6

perimeter P 24

total S 227.1

volume V 249.4

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cross-section	Kesim	Поперечное сечение
Prism	Prizma	Призма
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Pyramid	Piramida	Пирамида
Apex	Uchi (cho'qqi)	Вершина
Curved surface	Yon sirt	Боковая поверхность
Base area	Asos yuzasi	Площадь основания

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The cross-section of a prism is:

different at each cut

the same at every cut

always a circle

always a triangle

A cylinder has:

2 flat faces and 1 curved

3 flat faces

1 flat face

no faces

A cone has how many apexes?

0

1

2

4

The volume of any prism is:

base $\times$ height $\div 3$

base area $\times$ height

base + height

base area $\div$ height

A cylinder of radius 5 and height 10 holds (taking $\pi = 3.14$):

157

314

785

78.5

The curved surface of a cylinder unrolls into:

a circle

a rectangle

a triangle

a sector

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The cross-section of a prism is

ANSWER The same all along

2. The faces of a cylinder

ANSWER 2 flat and 1 curved

3. The faces of a cone

ANSWER 1 flat and 1 curved

4. The volume of a prism

ANSWER Base area $\times$ height

5. A prism of cross-section 12 cm^2 and length 10 cm

ANSWER 120 cm^3

6. The cross-section of a cuboid

ANSWER A rectangle

7. The point at the top of a cone

ANSWER The apex

Medium · the standard question

7 PROBLEMS

1. A prism of cross-section 20 cm^2 and length 8 cm

ANSWER 160 cm^3

2. A cylinder of $r = 5 \text{ cm}$ and $h = 10 \text{ cm}$, $\pi = 3.14$

ANSWER 785 cm^3

3. A cylinder of $r = 3 \text{ cm}$ and $h = 8 \text{ cm}$

ANSWER 226 cm^3

4. The area of a circle of radius 5 cm

ANSWER 78.5 cm^2

5. The curved surface of a cylinder $r = 5$, $h = 10$

ANSWER 314 cm^2

6. The faces of a pyramid on a square base

ANSWER 5

7. The net of a cone

ANSWER A circle and a sector

Hard · stretch and reason

7 PROBLEMS

1. A cylinder of $r = 7$ cm and $h = 10$ cm

ANSWER 1539 cm^3

2. The total surface of a cylinder $r = 5$, $h = 10$

ANSWER 471 cm^2

3. A triangular prism, cross-section base 6 height 4, length 10

ANSWER 120 cm^3

4. A pyramid on the same base and height as a prism holds

ANSWER A third as much

5. A prism of volume 240 cm^3 and length 8 cm: its cross-section

ANSWER 30 cm^2

6. A cylinder of volume 628 cm^3 and radius 5 cm: its height

ANSWER 8 cm

7. Which solid has a circular cross-section all along?

ANSWER The cylinder

07 Homework

Homework — 5 tasks

Write “taking $\pi = 3.14$ ” beside every answer that uses it.

1. A prism has cross-section 15 cm^2 and length 12 cm. Find its volume.
2. A cylinder has radius 4 cm and height 10 cm. Find its volume.
3. Find the curved surface area of that cylinder.
4. Count the faces, edges and vertices of a pentagonal prism.
5. Name the cross-section of a cuboid, a cylinder and a triangular prism.